Executive Summary

In this project, the main objective was to understand how feature engineering and preprocessing can improve the performance of machine learning model for predicting hotel booking cancellations. After completing all steps from baseline to final model, some important observations can be made.

First, what mattered most in this project was the quality of features rather than just using complex models. In the beginning, the baseline model was giving moderate performance because only basic features and minimal preprocessing was used. But after applying feature engineering techniques like creating new features (such as total nights, price per person, and special request rate), the model performance improved significantly. This shows that better representation of data helps model to learn patterns more effectively. Also, scaling of data (especially StandardScaler and RobustScaler) had strong impact in distance-based models like KNN.

Second, the factor that changed the performance ceiling was combination of preprocessing and feature construction. Just scaling alone improved the results slightly, but when new meaningful features was added, the model started capturing more real-world relationships. For example, features like total_nights and is_family gave better understanding of customer behavior. Also, removing redundant features using feature selection helped in reducing noise and improving stability of model. So, not only adding features but also selecting the right features is important.

Third, the most business-actionable features are those which directly relate to customer booking behavior. Features like lead_time, total_nights, adr (average daily rate), and number of special requests are very useful. For example, high lead_time bookings are more likely to cancel, so hotel can send reminders or offer small discounts. Similarly, customers with higher special requests may be more serious about booking, so they can be prioritized. Price per person also helps to understand if booking is expensive for customer, which can influence cancellation.

Overall, this project shows that feature engineering is very important step in machine learning pipeline. Instead of only focusing on model selection, working on data preprocessing and feature creation gives better and more stable results. It also makes model more interpretable, which is useful for business decisions. In conclusion, proper feature engineering and preprocessing helps in achieving better performance and also gives practical insights for real-world applications.